Estimated-Weight Infinitesimal Jackknife for Moment-Quadratic Estimators

June 20, 2026

Purpose

This note fixes the algebra for extending `robust_weighted_moment_ij` beyond complete all-ordinal ULS/DWLS/WLS. The target is the actual estimator whose second-stage weight is built from the data. The companion note `misspec_observed_bread.tex` derives the observed-Hessian bread and the complete all-ordinal diagonal-weight case. Here the goal is the general formula, the full-weight formula, and the implementation grid.

1 Moment estimator and scale

Let $\hat{\kappa}$ be a vector of stage-1 moments with casewise expansion

$$\hat{\kappa} = \kappa_0 + \frac{1}{N}\sum_{i=1}^{N} g_i + o_p(N^{-1/2}), \qquad \frac{1}{N}\sum_{i} g_i = O_p(N^{-1/2}).$$

For disjoint groups or blocks, stack the block moment vectors and set rows to zero outside the block. The model-implied moments are $\sigma(\theta)$, $d(\theta) = \sigma(\theta) - \hat{\kappa}$, and $\Delta = \partial\sigma/\partial\theta^{\mathsf{T}}$ is $m \times q$. Equality constraints are represented by a full-rank map K from reduced coordinates α to full θ, and $D = \Delta K$.

The fitted value minimizes

$$F_N(\theta) = \frac{1}{2} d(\theta)^{\mathsf{T}} \hat{W} d(\theta),$$

or the corresponding block sum. The reduced estimating equation is

$$U_N(\alpha) = D(\alpha)^{\mathsf{T}} \hat{W} d(\alpha) = 0.$$

At the pseudo-true value, let

$$A = \left.\frac{\partial U(\alpha)}{\partial \alpha^\mathsf{T}}\right|_{\alpha_*}$$

be the observed-Hessian bread. This includes the residual-contracted curvature term from $d(\theta)$; it is not the Gauss-Newton bread except under correct specification.

2 General estimated-weight IJ

Assume the weight has the first-order expansion

$$\hat{W} = W + \frac{1}{N}\sum_{i=1}^{N} \dot{W}_i + o_p(N^{-1/2}), \qquad \sum_i \dot{W}_i = O_p(N^{1/2}),$$

where $\dot{W}_i = \mathrm{IF}_i(\hat{W})$. Perturbing one case gives

$$\delta_i d = -g_i, \qquad \delta_i U = D^\mathsf{T}(\dot{W}_i d - W g_i).$$

Define the moment-space row contribution

$$v_i = W g_i - \dot{W}_i d. \tag{1}$$

Then

Theorem 1 (estimated-weight IJ). *For the data-weighted moment-quadratic estimator,*

$$\mathrm{IF}_i(\hat{\alpha}) = A^{-1} D^\mathsf{T} v_i,$$

up to an immaterial global sign convention for g_i. Hence

$$\widehat{\mathrm{Var}}(\hat{\theta}) = K A^{-1} \left\{ \frac{1}{N^2} \sum_{i=1}^{N} D^\mathsf{T} v_i v_i^\mathsf{T} D \right\} A^{-1} K^\mathsf{T}. \tag{2}$$

Proof. Linearize $0 = U_N(\hat{\alpha})$ around α_*. The data perturbation is $D^\mathsf{T}(\dot{W}_i d - W g_i)$ and the parameter perturbation is $A(\hat{\alpha} - \alpha_*)$. Solving for $\hat{\alpha} - \alpha_*$ yields $A^{-1} D^\mathsf{T}(W g_i - \dot{W}_i d)/N$ per case. Summing the outer products gives (2). □

Corollary 1 (fixed-weight reduction). *If $\dot{W}_i = 0$, $v_i = W g_i$. Since $\hat{\Gamma} = N^{-1} \sum_i g_i g_i^\mathsf{T}$, the IJ covariance in* (2) *reduces exactly to*

$$K A^{-1} D^\mathsf{T} W \hat{\Gamma} W D A^{-1} K^\mathsf{T} / N.$$

This is the primary regression gate: ULS and any caller-declared fixed weight must be machine-identical to the existing observed-bread sandwich.

3 Weight derivatives

3.1 Diagonal empirical weight

Let $\hat{W} = \operatorname{diag}(\hat{\gamma})^{-1}$ where $\hat{\gamma} = \operatorname{diag}(\hat{\Gamma})$. Then

$$\dot{W}_i = -\operatorname{diag}(\gamma^{-2} \odot \dot{\gamma}_i), \qquad v_i = W g_i + \operatorname{diag}(d \oslash \gamma^2)\dot{\gamma}_i.$$

In row form this is the implemented all-ordinal DWLS correction:

$$v_{ik} = g_{ik}/\gamma_k + d_k\, \dot{\gamma}_{ik}/\gamma_k^2.$$

For complete continuous DWLS, $\dot{\gamma}_i$ is the diagonal of (3). The implementation deliberately discards the off-diagonal empirical-Gamma influence before differentiating $\operatorname{diag}(\hat{\Gamma})^{-1}$; those off-diagonal channels belong to full WLS.

3.2 Full empirical weight

Let $\hat{W} = \hat{\Gamma}^{-1}$. Matrix inverse differentiation gives

$$\dot{W}_i = -W\dot{\Gamma}_i W, \qquad v_i = W g_i + W\dot{\Gamma}_i W d.$$

This is the full-WLS target. The required ingredient is the full matrix influence $\dot{\Gamma}_i$, not only its diagonal. A correct implementation must preserve symmetry and include all channels by which the stage-1 sandwich changes.

For complete continuous WLS/ADF, write $x_i = y_i - \bar{y}$ and $c_i = \operatorname{vech}(x_i x_i^\top) - \operatorname{vech}(S_N)$. The moment-influence row is $z_i = c_i$ without meanstructure and $z_i = (x_i^\top, c_i^\top)^\top$ with meanstructure, so

$$\hat{\Gamma} = \frac{1}{N}\sum_{j=1}^{N} z_j z_j^\top.$$

Under an infinitesimal case-weight perturbation at case i, set $h = x_i$. The exact finite-sample derivative is

$$\dot{\Gamma}_i = z_i z_i^\top - \hat{\Gamma} + \frac{1}{N}\sum_{j=1}^{N}\left\{\dot{z}_j(h) z_j^\top + z_j \dot{z}_j(h)^\top\right\}. \qquad (3)$$

If mean rows are present, $\dot{z}_{j,\mu}(h) = -h$. For the covariance component indexed by the lower-triangle pair (a, b),

$$\dot{z}_{j,ab}(h) = -(h_a x_{jb} + x_{ja} h_b).$$

The derivative of the subtracted covariance S_N is constant across j and therefore drops from (3) because $N^{-1}\sum_j z_j = 0$. The implemented continuous WLS correction uses $d^{\mathsf{T}} W \dot{\Gamma}_i W$ in row form and is regression-tested against central finite differences of the contaminated empirical weight $\hat{W}_\epsilon = \hat{\Gamma}_\epsilon^{-1}$, including the stacked mean–covariance cross-block.

3.3 DLS and mixed weights

For a Browne-style mixture

$$\hat{W} = \left((1-a)\hat{\Gamma}_0 + a\hat{\Gamma}_1\right)^{-1},$$

write $M = (1-a)\Gamma_0 + a\Gamma_1$. Then

$$\dot{W}_i = -W\{(1-a)\dot{\Gamma}_{0i} + a\dot{\Gamma}_{1i}\}W, \qquad v_i = Wg_i + W\{(1-a)\dot{\Gamma}_{0i} + a\dot{\Gamma}_{1i}\}Wd.$$

If Γ_0 is model-implied and depends only on θ, its derivative belongs in the bread. If Γ_0 is built from saturated sample moments, as in two-stage FIML's normal-theory Stage-2 weight, its casewise derivative is a weight-IJ term. For complete-data continuous DLS with caller-fixed a, $\Gamma_0 = \Gamma_{\mathrm{NT}}(\hat{S})$ and $\Gamma_1 = \hat{\Gamma}_{\mathrm{ADF}}$. The implemented row correction is the same convex mixture of the normal-theory and dense empirical-Gamma IF channels, evaluated with the DLS weight. In meanstructure layouts the normal-theory side is $\mathrm{blockdiag}(\hat{S}, \Gamma_{\mathrm{NT}}(\hat{S}))$ and the empirical side is the full stacked Browne $\hat{\Gamma}_{\mathrm{ADF}}$, including the mean–covariance cross-block. The scalar a is treated as fixed; an empirical-Bayes rule for selecting a would add a separate scalar IF term.

4 Stage-1 sandwich influence

Many magmaan moments are themselves M-estimators. A common form is

$$\hat{\Gamma} = \hat{A}^{-1}\hat{V}\hat{A}^{-\mathsf{T}}, \qquad \hat{V} = N^{-1}\sum_i s_i s_i^{\mathsf{T}}, \qquad \hat{A} = N^{-1}\sum_i b_i.$$

The data-direct influence at fixed stage-1 parameter is

$$\dot{\Gamma}_i^{\mathrm{direct}} = A^{-1}(s_i s_i^{\mathsf{T}} - V)A^{-\mathsf{T}} - A^{-1}(b_i - A)\Gamma - \Gamma(b_i - A)^{\mathsf{T}}A^{-\mathsf{T}}.$$

If Γ also changes through the movement of the stage-1 parameter κ, add

$$\dot{\Gamma}_i^{\kappa} = \left[\frac{\partial \Gamma}{\partial \kappa}\right] g_i.$$

The all-ordinal path now computes $\dot{\Gamma}_i^{\mathrm{direct}}$ from integer data and the diagonal of $\dot{\Gamma}_i^{\kappa}$ by finite differences for DWLS. Full WLS uses the dense matrix $\dot{\Gamma}_i^{\mathrm{direct}} + \dot{\Gamma}_i^{\kappa}$ with the row correction $d^{\mathsf{T}} W \dot{\Gamma}_i W$.

5 Implementation grid

Complete all-ordinal. ULS, DWLS, and WLS are implemented for complete integer data. ULS is covered by Corollary 1. DWLS uses the diagonal formula above. Full WLS uses the full $\dot{\Gamma}_i$ matrix in the dense inverse-weight derivative. The dense data-direct channel is gated against case-weight finite differences of the full NACOV matrix; its diagonal extraction is also checked against the DWLS helper. A deterministic resampling-jackknife fixture and misspecification simulation remain useful higher-level checks.

Complete mixed ordinal/polyserial. The same equations apply, but g_i and $\dot{\Gamma}_i$ must cover continuous means, continuous covariances, ordinal thresholds, polychorics, and polyserials in the exact mixed moment order. Fixed-weight ULS is implemented: it only needs casewise g_i and reduces to the observed-bread sandwich. DWLS is also implemented for ordinary complete-data ML/polyserial mixed stats: the diagonal $\dot{\Gamma}_i$ combines the data-direct sandwich channel at fixed mixed kappa with a finite-difference $\partial\, \mathrm{diag}(\hat{\Gamma})/\partial\kappa$ channel for thresholds, negative means, variances, polychorics, polyserial covariances, and Pearson covariance rows. Full WLS uses the dense mixed $\dot{\Gamma}_i$ with the same data-direct and finite-difference channels, and its diagonal extraction is checked against the DWLS helper.

Complete continuous LS. For ULS with a fixed identity weight, the IJ is only a wrapper around the observed-bread sandwich using raw moment influence rows. This slice has landed as the fixed-weight continuous-LS IJ helper. The sample-built GLS normal-theory-weight slice has also landed: this is a moment-quadratic estimated-weight correction, not ordinary ML/FIML robust-score normal-theory inference. The complete-data WLS/ADF empirical-weight slice has also landed, using (3) for the full dense $\dot{\Gamma}_i$ and a finite-difference contamination gate for both covariance-only and meanstructure layouts. Complete-data DWLS has landed as the diagonal extraction of the same empirical-Gamma influence, gated by case-weight finite differences of $\mathrm{diag}(\hat{\Gamma})^{-1}$ in covariance-only and meanstructure layouts. Complete-data DLS with fixed mixing scalar has also landed as the convex mixture of the sample-built normal-theory and dense empirical-Gamma influence channels, with finite-difference gates on the mixed inverse weight in covariance-only and meanstructure layouts. The raw mean/covariance g_i is analytic; EB-DLS scalar-selection uncertainty remains separate from the fixed-a weight derivative.

Two-stage FIML (ML2S). Stage 1 supplies saturated EM moments and their ACOV. The observed-bread single-model regime has landed as `TwoStageBread::Observed`: it changes the Stage-2 bread while leaving the Stage-1 ACOV and the current fixed-weight robust sandwich unchanged. Complete-data gates reduce the NT path to unstructured observed-bread robust SEM and the ADF path to observed-bread `robust_continuous_ls`. The remaining IJ object is casewise saturated-moment influence under the missingness pattern. For `TwoStageWeight::Nt`, the weight still depends on saturated $\hat{\Sigma}$ through $\Gamma_{\mathrm{NT}}(\hat{\Sigma})$, so the target estimator has a normal-theory weight derivative. For `Dwls`, `Adf`, and `Dls`, use the diagonal, full, and mixed formulas above with Γ_{FIML}.

MCAR and pairwise-missing ordinal/polyserial. The current primitive assumes disjoint blocks: it sums $V_b^{\mathsf{T}} V_b$ inside each block. Pairwise missingness has overlapping supports. The correct object is one case-aligned sparse row vector over the full moment stack, with zeros for unobserved moment components and the right N/n_{jk} scaling for each pair-local estimator. This probably needs a second primitive. An extension of the current IJ block would need case ids and per-moment support scaling. The gates are complete-data reduction, MCAR simulation against a case-resampling jackknife, and equality of pairwise-overlap covariance to the existing NACOV.

6 Verification standard

Every adapter gets the same four checks before it is trusted:

1. fixed-weight reduction to `robust_weighted_moments`;
2. finite-difference or case-weight derivative check for each new $\dot{\Gamma}_i$ channel;
3. resampling-jackknife agreement on a small deterministic fixture;
4. simulation calibration under the misspecified model that motivated the method, with ULS/fixed-weight cells used as the negative control.

References

[1] Hall, A. R., and Inoue, A. (2003). The large sample behaviour of the generalized method of moments estimator in misspecified models. *Journal of Econometrics*, 114(2), 361–394. doi:10.1016/S0304-4076(03)00075-0.